USB4 2.0 ENGINEERING CHANGE NOTICE FORM

USB4 2.0 ECN - Gen 4 CLx Exit in combination with Recovery Enabled timeout Title: TX ISI MARGIN Limit Adjustment Applied to: USB4 Specification Version 2.0

Brief description of the functional changes :
Change Gen4 TX_ISI_MARGIN limit from 11.5dB to 11dB

Benefits as a result of the changes:
Adjusting the spec limit for better considering the penalty associated with the lab measurement setup that impacts the result more than originally predicted

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
None

Actual Change

(a). Table 3-22

From Text:

Table 3-22. Transmitter Specifications for Gen 4 (at TP2)

Symbol	Description	Min	Max	Units	Comments
UI	Minimum Unit Interval	39.0508	39.0742	ps	Corresponding to the Gen 4 baseline Baud rate of 25.6GB with an uncertainty range of -300 ppm to 300 ppm. See Notes 1, 2.
V_SWING	Peak differential voltage swing	390	500	mV	See Note 9 and Section 3.2.3.2.
TX_LEVELS_MISMATCH	Levels separation mismatch ratio	0.975			See Note 9 and Section 3.2.3.3.
TX_SNDR	Signal to Noise and Distortion Ratio	32.5		dB	See Note 9 and Section 3.2.3.4.
TX_ISI_MARGIN	Signal to Residual ISI Ratio	11.5		dB	See Note 9 and Section 3.2.3.5.

To Text:

Table 3-22. Transmitter Specifications for Gen 4 (at TP2)

Symbol	Description	Min	Max	Units	Comments
UI	Minimum Unit Interval	39.0508	39.0742	ps	Corresponding to the Gen 4 baseline Baud rate of 25.6GB with an uncertainty range of -300 ppm to 300 ppm. See Notes 1, 2.
V_SWING	Peak differential voltage swing	390	500	mV	See Note 9 and Section 3.2.3.2.
TX_LEVELS_MISMATCH	Levels separation mismatch ratio	0.975			See Note 9 and Section 3.2.3.3.
TX_SNDR	Signal to Noise and Distortion Ratio	32.5		dB	See Note 9 and Section 3.2.3.4.
TX_ISI_MARGIN	Signal to Residual ISI Ratio	~~11.5~~11		dB	See Note 9 and Section 3.2.3.5.

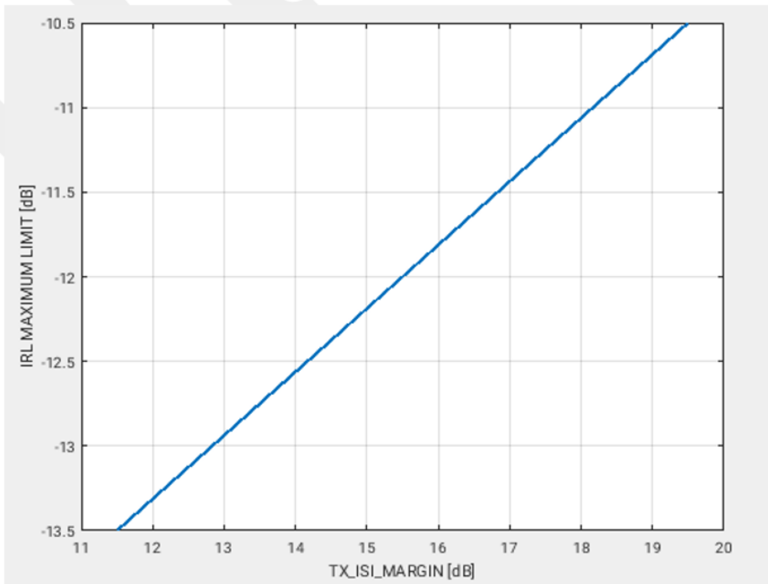
(b). Section 3.2.3.5

From Text:

The TX_IRL maximum limit is a function of the measured TX_ISI_MARGIN, as specified in Section 3.2.3.5:

$$IRL \leq -13.5 + (TX\_ISI\_MARGIN \ - 11.5) \times 0.375$$

Figure 3-30. IRL Maximum Limit as a Function of TX_ISI_MARGIN



To Text:

The TX_IRL maximum limit is a function of the measured TX_ISI_MARGIN, as specified in Section 3.2.3.5:

$$IRL \leq -13.5 + (TX\_ISI\_MARGIN \; - \cancel{11.5}11) \times 0.375$$

Figure 3-30. IRL Maximum Limit as a Function of TX_ISI_MARGIN